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1 Goals of Usability

1.1 Learnability

Definition:

Learnability refers to how easy it is for users to accomplish basic tasks the first time they encounter the design.

Importance:

- Users should be able to quickly learn how to use the system without extensive training.
- Improves the initial user experience and reduces frustration.

1.2 Memorability

Definition:

Memorability is the ease with which users can remember how to use the system after a period of not using it.

Importance:

- Ensures that returning users can quickly re-establish proficiency.
- Reduces the need for re-learning, thus enhancing user satisfaction.

1.3 Efficiency

Definition:

Efficiency refers to how quickly users can perform tasks once they have learned the design.

Importance:

- Users should be able to perform tasks in a timely manner.
- Enhances productivity and user satisfaction by reducing the time and effort required to complete tasks.

1.4 Error Tolerance

Definition:

Error tolerance is the ability of the system to help users avoid errors and recover from them easily when they do occur.

Importance:

- Minimizes the negative impact of errors on the user experience.
- Provides users with clear, helpful feedback to correct mistakes.

1.5 Satisfaction

Definition:

Satisfaction refers to how pleasant and satisfying the system is to use.

Importance:

- Ensures a positive emotional response from users.
- Encourages continued use and adoption of the system.

2 User-Centered Design

Definition

User-centered design (UCD) is a design philosophy and process that places the needs, wants, and limitations of end users at the forefront of each stage of the design process.

Key Principles

- **User Involvement:** Involve users throughout the design process to gather their feedback and understand their needs.
- **Iterative Design:** Continuously refine and improve the design based on user feedback and testing.
- **Understanding Users:** Conduct user research to gain deep insights into user behavior, needs, and contexts.
- **Design for the User:** Focus on creating solutions that are tailored to the specific needs and preferences of the target users.
- **Evaluation and Testing:** Regularly test the design with real users to identify issues and gather feedback for improvement.

Process**1. Research:**

- Conduct user interviews, surveys, and observations to understand user needs and behaviors.

- Create user personas to represent different user types and their goals.

2. Design:

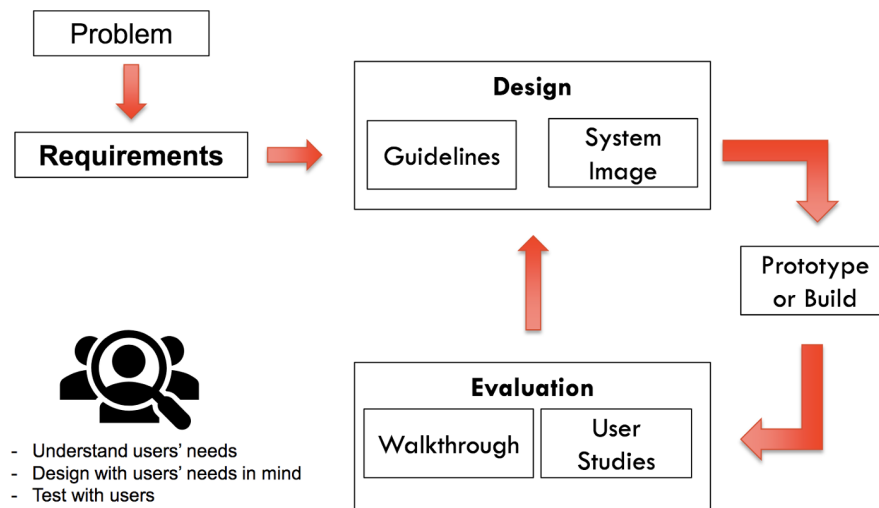
- Develop wireframes, prototypes, and design mockups based on research insights.
- Ensure designs are accessible and inclusive, catering to a diverse user base.

3. Testing:

- Conduct usability testing to evaluate the effectiveness of the design.
- Gather feedback and identify areas for improvement.

4. Iteration:

- Refine the design based on testing results and user feedback.
- Repeat the process until the design meets user needs and goals effectively.



Conclusion

Balancing the five goals of usability—learnability, memorability, efficiency, error tolerance, and satisfaction—is crucial for creating effective and user-friendly systems. User-centered design is an iterative and inclusive approach that ensures the end product aligns closely with the needs and expectations of its users, leading to higher user satisfaction and success.

3 PACT Framework

3.1 Introduction

The PACT framework is a tool used in user-centered design to understand and analyze four key elements: People, Activities, Contexts, and Technologies. This framework helps designers create more effective, user-friendly systems by considering all relevant factors that influence the user experience.

3.2 Components of PACT

3.2.1 People

Definition: This element focuses on understanding the users of the system, their characteristics, needs, and capabilities.

Key Considerations:

- **Demographics:** Age, gender, education level, cultural background, etc.
- **Physical Abilities:** Any physical limitations or disabilities that might affect how users interact with the system.
- **Cognitive Abilities:** Memory, problem-solving skills, attention span, etc.
- **Preferences and Expectations:** User preferences, past experiences with similar systems, and expectations.

Example:

- Designing a mobile application for elderly users may require larger text sizes and simplified interfaces to accommodate their physical and cognitive abilities.

3.2.2 Activities

Definition: This element examines the tasks and activities that users will perform with the system.

Key Considerations:

- **Task Complexity:** The difficulty and number of steps required to complete a task.
- **Frequency:** How often users perform certain tasks.
- **Duration:** The time it takes to complete tasks.
- **Collaboration:** Whether tasks are performed individually or require collaboration with others.

Example:

- A project management tool must support both individual task tracking and collaborative features such as shared calendars and communication channels.

3.2.3 Contexts

Definition: This element considers the various contexts in which users interact with the system, including physical, social, and organizational environments.

Key Considerations:

- **Physical Environment:** The physical settings where the system will be used (e.g., at home, in the office, outdoors).
- **Social Environment:** The social dynamics and interactions that might influence how the system is used (e.g., peer influence, team collaboration).
- **Organizational Environment:** The organizational structures, policies, and cultures that impact system use (e.g., workplace policies, management support).

Example:

- A navigation app must be designed for use in various physical environments, such as walking in a city, driving a car, or using public transport.

3.2.4 Technologies

Definition: This element focuses on the technologies that will be used to support the interactions between users and the system.

Key Considerations:

- **Hardware:** The physical devices used (e.g., smartphones, tablets, desktop computers).
- **Software:** The applications and systems that run on the hardware (e.g., operating systems, web browsers, mobile apps).
- **Input/Output:** The methods by which users interact with the system (e.g., touch screens, keyboards, voice commands).
- **Connectivity:** The network and communication technologies required (e.g., Wi-Fi, Bluetooth, cellular networks).

Example:

- Designing an e-learning platform requires consideration of various devices (e.g., desktops for detailed work, tablets for mobility) and input methods (e.g., touch, keyboard).

3.3 Conclusion

The PACT framework provides a comprehensive approach to understanding the critical aspects of user-centered design. By analyzing People, Activities, Contexts, and Technologies, designers can create more effective and user-friendly systems. This holistic approach ensures that all relevant factors are considered, leading to better user experiences and more successful products.

4 Another Framework for Usability

C ustomers	Clients – the users of the process or system • Drivers, passengers
A ctors	People who take part in the implementation of the changes in the process or system. • Developers
T ransformation	Change that the system or process brings • A mobile application that...
W orldview	The big picture • Increase transportation options for passengers and income for drivers.
O wner	Owner of the problem. • CEO/shareholders of UBER
E nvironment Constraints	Key constraints outside the system boundary that are significant to the system. • Local laws of the country, road rules, regulation problems

5 Persona and Its Role in Usability Process

5.1 Persona

Definition:

A persona is a fictional character that represents a user type within a targeted demographic, attitude, and behavior set. Personas are used in user-centered design to help understand the users' needs, experiences, behaviors, and goals.

Components of a Persona:

- **Name and Demographics:** A fictional name, age, gender, occupation, and other demographic details.
- **Background and Context:** Information about the persona's life, work, and personal context.
- **Goals and Needs:** The persona's primary goals and needs when using the product or system.
- **Pain Points:** Challenges and frustrations the persona faces in their interactions.
- **Behaviors and Attitudes:** The persona's typical behaviors, attitudes, and motivations.

5.2 Role in Usability Process:

- **Research Phase:** Personas are developed based on user research, including interviews, surveys, and observations.
- **Design Phase:** Personas guide the design decisions, ensuring that the product meets the needs of its users.
- **Testing Phase:** Personas are used to create scenarios for usability testing, helping to evaluate whether the design works well for different user types.

6 Card Sorting and Its Types

6.1 Card Sorting

Definition:

Card sorting is a user research technique used to help design or evaluate the information architecture of a product. In a card sorting session, participants organize topics into categories that make sense to them and may also help label these groups.

6.2 Types of Card Sorting

6.2.1 Open Card Sorting

Description: Participants are given a set of cards with content or topics and are asked to organize them into groups that make sense to them. They then label each group with a name they think is appropriate.

Advantages:

- Provides insight into how users naturally categorize information.
- Helps identify the users' mental models and terminology.

Disadvantages:

- Can result in a wide variety of categories, making analysis more complex.

6.2.2 Closed Card Sorting

Description: Participants are given a set of cards and a predefined set of category names. They are asked to sort the cards into these predefined categories.

Advantages:

- Easier to analyze as the categories are predefined.
- Useful for testing existing category structures.

Disadvantages:

- Does not provide insights into users' natural categorization or terminology.

6.2.3 Hybrid Card Sorting

Description: Combines elements of both open and closed card sorting. Participants can sort cards into predefined categories but also have the option to create new categories if needed.

Advantages:

- Provides a balance between user-driven categorization and predefined structures.
- Allows for flexibility in understanding user preferences.

Disadvantages:

- Can be more complex to analyze due to the mix of predefined and user-created categories.

6.3 Relative Information

Purpose of Card Sorting:

- To understand how users group and label information.
- To inform the information architecture and navigation structure of a website or application.
- To ensure that the organization of information aligns with users' expectations and mental models.

6.4 Conducting a Card Sorting Session:

1. **Preparation:** Define the goal of the session, select participants, and prepare the cards with content or topics.
2. **Session:** Provide participants with instructions, conduct the sorting activity, and collect the results.
3. **Analysis:** Analyze the grouped data to identify patterns and insights, and use these to inform design decisions.

7 Summary of Testing Methods

7.1 Moderated vs. Unmoderated Testing

7.1.1 Moderated Testing

Definition: A testing method where a facilitator is present during the session to guide participants, answer questions, and observe behaviors.

Advantages:

- Facilitator can provide immediate clarification and support.
- Allows for real-time observation of participant behavior and non-verbal cues.
- Flexibility to probe deeper into specific issues as they arise.

Disadvantages:

- More resource-intensive, requiring a facilitator and potentially more setup.
- Participants may feel observed, which can influence their behavior.

7.1.2 Unmoderated Testing

Definition: A testing method where participants complete tasks without a facilitator present, typically using automated tools to record their actions and responses.

Advantages:

- More cost-effective and scalable, allowing for larger sample sizes.
- Participants may behave more naturally without a facilitator present.
- Can be conducted asynchronously, offering flexibility for participants.

Disadvantages:

- Lack of immediate support and clarification for participants.
- Limited ability to observe non-verbal cues and behaviors.
- Less flexibility to explore issues that arise during the session.

7.2 Individual vs. Group Testing

7.2.1 Individual Testing

Definition: A testing method where a single participant is tested at a time.

Advantages:

- Provides detailed insights into individual user behaviors and preferences.
- Reduces the influence of group dynamics on individual responses.
- Easier to manage and analyze data from individual sessions.

Disadvantages:

- Can be time-consuming and resource-intensive for large sample sizes.
- Lacks the opportunity for collaborative feedback and discussion.

7.2.2 Group Testing

Definition: A testing method where multiple participants are tested together, often in a workshop or focus group setting.

Advantages:

- Encourages discussion and collaborative feedback.
- Can quickly gather diverse perspectives and identify common issues.
- Efficient for collecting data from multiple participants at once.

Disadvantages:

- Group dynamics can influence individual responses, potentially leading to conformity.
- More challenging to manage and analyze data from group sessions.
- Difficult to observe individual behaviors and preferences in detail.

7.3 Physical vs. Remote Testing

7.3.1 Physical Testing

Definition: A testing method conducted in person, with participants and facilitators present in the same physical location.

Advantages:

- Facilitators can observe non-verbal cues and behaviors directly.
- Easier to build rapport and trust with participants.
- Immediate support and clarification are available for participants.

Disadvantages:

- More logistical challenges and costs related to travel, venue, and setup.
- Limited to participants who can physically attend the session.

7.3.2 Remote Testing

Definition: A testing method conducted online, with participants and facilitators in different locations, using digital tools for communication and observation.

Advantages:

- More flexible and cost-effective, allowing for a wider geographical reach.
- Participants can take part from the comfort of their own environment.
- Easier to schedule sessions and accommodate participants' availability.

Disadvantages:

- Limited ability to observe non-verbal cues and behaviors.
- Potential technical issues related to internet connectivity and software.
- Less personal interaction, which can affect the rapport with participants.

8 Information Architecture

8.1 Definition

Information architecture (IA) is the practice of organizing, structuring, and labeling content in an effective and sustainable way. The goal is to help users find information and complete tasks. IA involves creating a blueprint for how information is organized and presented in a digital environment.

8.2 Components of Information Architecture

8.2.1 Organization and Labeling System

Definition: The organization and labeling system defines how content is grouped and categorized, as well as how it is named and represented.

Organization Systems:

- **Hierarchical:** Content is organized in a tree-like structure with parent and child relationships. It helps users understand the context and find specific information.
- **Sequential:** Content is organized in a specific order, guiding users through a step-by-step process. It is useful for task-oriented processes.
- **Matrix:** Content is organized in a grid or matrix, allowing users to navigate based on multiple criteria or attributes. It supports flexible access to information.
- **Database-Oriented:** Content is organized in a database structure, where information can be accessed through queries and relationships. It is useful for complex and dynamic data sets.

Labeling Systems:

- **Descriptive Labels:** Provide clear and concise descriptions of content, helping users understand what each section or item contains.
- **Contextual Labels:** Provide additional context to help users understand the relevance and purpose of the content.
- **Conventional Labels:** Use commonly understood terms and conventions to ensure users can easily recognize and navigate the content.

8.2.2 Navigation System

Definition: The navigation system defines how users move through the content and find the information they need. It includes the design and structure of menus, links, and other navigational elements.

Types of Navigation Systems:

- **Global Navigation:** Provides access to the main sections of a website or application. It is typically consistent across all pages.
- **Local Navigation:** Provides access to sub-sections within a specific section of the website or application. It helps users navigate within a particular area.
- **Contextual Navigation:** Provides links to related content based on the current context. It helps users discover additional relevant information.
- **Breadcrumb Navigation:** Shows the user's current location within the hierarchy and provides links to higher levels. It helps users understand their context and navigate back.
- **Faceted Navigation:** Allows users to filter and refine content based on multiple attributes or facets. It is useful for large and complex data sets.

8.2.3 Search System

Definition: The search system allows users to find specific information by entering queries. It includes the design and functionality of search features, as well as how search results are presented.

Components of Search Systems:

- **Search Interface:** The user interface for entering search queries and displaying results. It should be intuitive and easy to use.
- **Search Algorithms:** The logic and algorithms used to process search queries and retrieve relevant results. It should be accurate and efficient.
- **Search Filters:** Options for refining and narrowing down search results based on various criteria. It helps users find the most relevant information.
- **Search Results Presentation:** How search results are displayed, including the layout, ranking, and relevance indicators. It should be clear and informative.

8.3 Importance of Information Architecture

- **Usability:** A well-organized and labeled information architecture enhances the usability of a website or application, making it easier for users to find what they need.
- **User Satisfaction:** Effective navigation and search systems improve the user experience, leading to higher satisfaction and engagement.
- **Efficiency:** Efficient information retrieval reduces the time and effort required for users to complete tasks, increasing productivity.
- **Consistency:** Consistent organization, labeling, and navigation systems provide a coherent and predictable user experience across different sections of the website or application.

9 Sitemap and Wireframe

9.1 Sitemap

Definition:

A sitemap is a visual representation of the structure and hierarchy of a website. It outlines the organization of content and the relationships between different pages and sections.

Key Components:

- **Home Page:** The main entry point of the website.
- **Main Sections:** Primary categories or areas of the website, often represented as top-level pages.
- **Subsections:** Subcategories or specific content areas within the main sections, represented as nested pages.
- **Links:** Connections between pages, showing how users can navigate from one page to another.

Purpose:

- Provides a clear overview of the website's structure.
- Helps identify the most important content and its placement.
- Facilitates communication between designers, developers, and stakeholders.
- Aids in planning navigation and ensuring a logical flow of information.

Example:

- A simple sitemap for an e-commerce website might include the home page, product categories, individual product pages, a shopping cart, and a check-out page.

9.2 Wireframe

Definition:

A wireframe is a low-fidelity visual representation of a web page or application's layout and structure. It outlines the placement of elements such as headers, footers, content areas, navigation menus, and interactive components.

Key Components:

- **Layout:** The arrangement of elements on the page, including the header, footer, sidebars, and main content area.
- **Navigation:** Placement and structure of menus, links, and buttons for navigating the website or application.
- **Content Blocks:** Areas designated for text, images, videos, and other content types.
- **Interactive Elements:** Buttons, forms, and other elements that require user interaction.

Purpose:

- Provides a visual guide for the design and development process.
- Helps identify the most important elements and their placement.
- Facilitates communication between designers, developers, and stakeholders.
- Aids in planning the user experience and ensuring a logical flow of information.

9.3 Types of Wireframes

9.3.1 Low-Fidelity Wireframes

Description: Basic sketches or simple digital drawings that focus on the layout and structure without detailed design elements.

Characteristics:

- Simple and quick to create.
- Focus on the overall structure and layout.

- Lack detailed design elements, colors, and images.

Purpose:

- Used in the early stages of the design process to brainstorm and explore layout options.
- Facilitates quick feedback and iteration.

9.3.2 High-Fidelity Wireframes

Description: Detailed representations that closely resemble the final design, including colors, typography, and images.

Characteristics:

- Highly detailed and polished.
- Include actual content and design elements.
- Closely resemble the final product.

Purpose:

- Used to finalize the design and layout.
- Facilitates detailed usability testing and stakeholder approval.

10 Usability Evaluation Methods

10.1 Cognitive Walkthrough

Definition:

A Cognitive Walkthrough is a usability evaluation method where evaluators go through a task step-by-step from the user's perspective to identify usability issues. The focus is on understanding the cognitive processes required to complete the task.

Key Steps:

1. **Define User Goals:** Identify the specific tasks users want to accomplish.
2. **Create Task Scenarios:** Develop realistic scenarios that users might encounter.
3. **Identify Action Sequence:** List the steps users need to take to complete the task.
4. **Evaluate Each Step:** At each step, evaluators ask:
 - Will the user try to achieve the right effect?

- Will the user notice that the correct action is available?
- Will the user associate the correct action with the effect they are trying to achieve?
- If the correct action is performed, will the user see that progress is being made toward their goal?

Advantages:

- Focuses on user goals and cognitive processes.
- Can be conducted with minimal resources.
- Identifies usability issues early in the design process.

Disadvantages:

- Requires detailed task scenarios and action sequences.
- May miss issues that only occur in real-world usage.

10.2 Think Aloud Test

Definition:

The Think Aloud Test is a usability testing method where participants verbalize their thoughts, feelings, and actions while interacting with a system. This method helps evaluators understand the user's cognitive processes and identify usability issues.

Key Steps:

1. **Recruit Participants:** Select users who represent the target audience.
2. **Prepare Tasks:** Develop a set of tasks for participants to complete.
3. **Conduct the Test:** Ask participants to think aloud while performing the tasks. Encourage them to verbalize their thoughts, actions, and any difficulties they encounter.
4. **Record Observations:** Take notes or record the session for later analysis.
5. **Analyze Results:** Review the verbalizations and identify usability issues and areas for improvement.

Advantages:

- Provides insights into the user's thought processes and decision-making.
- Helps identify specific areas of confusion or frustration.
- Can reveal unexpected user behaviors and issues.

Disadvantages:

- Participants may feel self-conscious or find it difficult to verbalize their thoughts.
- The presence of an observer may influence participant behavior.
- Analysis of verbal data can be time-consuming.

10.3 Guerrilla Test

Definition:

The Guerrilla Test is a quick and informal usability testing method conducted in public places with randomly selected participants. It is often used to gather quick feedback on design concepts or prototypes.

Key Steps:

1. **Choose a Location:** Select a public place where potential users are likely to be found (e.g., cafes, parks, malls).
2. **Prepare Materials:** Bring the design concept or prototype to be tested, along with any necessary recording equipment.
3. **Recruit Participants:** Approach people and ask if they are willing to participate in a short usability test.
4. **Conduct the Test:** Ask participants to complete a task using the design concept or prototype and gather their feedback.
5. **Record Observations:** Take notes or record the session for later analysis.
6. **Analyze Results:** Review the feedback and identify usability issues and areas for improvement.

Advantages:

- Quick and cost-effective way to gather user feedback.
- Provides insights from a diverse range of users.
- Can be conducted with minimal preparation and resources.

Disadvantages:

- May not provide in-depth insights due to the brief nature of the interactions.
- Participants may not represent the target user population accurately.
- Limited control over the testing environment.

11 Selected Norman's Design Principles

Norman's design principles are foundational guidelines for creating user-friendly and intuitive interfaces. Here, we focus on six key principles: visibility, affordance, feedback, mapping, constraints, and consistency.

11.1 Visibility of System Status

Definition:

The system should always keep users informed about what is going on through appropriate feedback within a reasonable amount of time.

Key Points:

- Users should know the current state of the system at any given time.
- Feedback should be immediate and clearly visible.

Example:

- A loading spinner indicating that a webpage is still loading.
- A progress bar showing the status of a file upload.

11.2 Affordance

Definition:

Affordance refers to the perceived and actual properties of an object that determine how it can be used. It suggests how an object should be used.

Key Points:

- Design elements should suggest their functionality.
- Users should intuitively understand how to interact with the interface.

Example:

- A button that looks clickable and invites the user to press it.
- A door handle that suggests pulling or pushing.

11.3 Feedback

Definition:

Feedback is about providing users with information about what action has been done and what has been accomplished, ensuring that the user knows their action was successful.

Key Points:

- Users should receive immediate feedback for their actions.
- Feedback should be clear and informative.

Example:

- A sound or visual notification confirming that an email has been sent.
- Highlighting a form field in red if the input is incorrect.

11.4 Mapping

Definition:

Mapping refers to the relationship between controls and their effects in the world. Good mapping ensures that the controls are logically placed and easy to use.

Key Points:

- The layout of controls should correspond to the intended actions.
- Users should easily understand the relationship between actions and their results.

Example:

- Stove controls that are laid out in the same pattern as the burners.
- Steering a car: turning the steering wheel to the right results in the car turning right.

11.5 Constraints

Definition:

Constraints limit the possible actions that can be performed, helping prevent user errors by restricting the actions that can be taken.

Key Points:

- Design should limit the number of available actions to prevent errors.
- Constraints guide users towards correct usage.

Example:

- Grayed-out buttons that are inactive and cannot be clicked.
- A USB plug that can only be inserted one way.

11.6 Consistency

Definition:

Consistency refers to designing interfaces to have similar operations and use similar elements for achieving similar tasks. This helps users understand and predict the interface's behavior.

Key Points:

- Maintain uniformity in design elements and behaviors.
- Use familiar conventions to meet user expectations.

Example:

- Using the same icons and terminology throughout an application.
- Consistent placement of navigation menus across different pages.

12 Design Principle/Guideline/Rules

Design principle: Provide plain-language error messages to pinpoint problems and likely solutions.

Design Guideline: Write large-lettered, jargon-free text in web-safe font. Use short sentences and draw users' attention to causes and remedies.

Design rule: Use 20-pt, black Georgia on lavender background (#e6e6fa Hex). Put instructions in bold.

13 Accessibility and POUR Principles

13.1 Accessibility

Definition

: Accessibility refers to the design of products, devices, services, or environments so that they are usable by people with disabilities. The goal is to create inclusive designs that accommodate the needs of all users.

Key Components

:

- **Visual Accessibility:** Ensuring that content is perceivable by users with visual impairments (e.g., providing text alternatives for images, ensuring sufficient color contrast).
- **Auditory Accessibility:** Making content accessible to users with hearing impairments (e.g., providing captions for videos, transcripts for audio).
- **Motor Accessibility:** Designing interfaces that are usable by individuals with motor impairments (e.g., keyboard navigation, voice commands).
- **Cognitive Accessibility:** Creating designs that are understandable and usable by individuals with cognitive impairments (e.g., clear and simple language, consistent navigation).

13.2 POUR Principles

POUR

is an acronym that stands for Perceivable, Operable, Understandable, and Robust. These principles provide a framework for creating accessible digital content.

13.2.1 Perceivable

Definition: Information and user interface components must be presented to users in ways they can perceive.

Key Points :

- Provide text alternatives for non-text content (e.g., images, videos).
- Ensure content is adaptable and distinguishable (e.g., use sufficient color contrast, enable text resizing).

13.2.2 Operable

Definition : User interface components and navigation must be operable by all users.

Key Points :

- Make all functionality available from a keyboard.
- Provide users enough time to read and use content.
- Avoid designing content in a way that is known to cause seizures.
- Provide ways to help users navigate, find content, and determine where they are.

13.2.3 Understandable

Definition : Information and the operation of user interface components must be understandable.

Key Points :

- Make text content readable and understandable.
- Make web pages appear and operate in predictable ways.
- Help users avoid and correct mistakes.

13.2.4 Robust

Definition : Content must be robust enough to be interpreted reliably by a wide variety of user agents, including assistive technologies.

Key Points :

- Maximize compatibility with current and future user agents, including assistive technologies.

14 Nielsen's 10 Usability Heuristics

Jakob Nielsen's 10 usability heuristics are general principles for user interface design. They are called "heuristics" because they are broad rules of thumb and not specific usability guidelines.

1. Visibility of System Status

Definition: The system should always keep users informed about what is going on, through appropriate feedback within a reasonable time.

Example:

- A progress bar indicating the status of a file upload.

2. Match Between System and the Real World

Definition: The system should speak the users' language, with words, phrases, and concepts familiar to the user, rather than system-oriented terms.

Example:

- Using a shopping cart icon for the e-commerce feature.

3. User Control and Freedom

Definition: Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue.

Example:

- An "Undo" button in a text editor.

4. Consistency and Standards

Definition: Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.

Example:

- Consistent use of icons and terminology throughout the application.

5. Error Prevention

Definition: Even better than good error messages is a careful design which prevents a problem from occurring in the first place.

Example:

- Disabling the "Submit" button until all required fields are filled out correctly.

6. Recognition Rather Than Recall

Definition: Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another.

Example:

- Providing a dropdown menu with options rather than requiring users to remember and type in options.

7. Flexibility and Efficiency of Use

Definition: Accelerators—unseen by the novice user—may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users.

Example:

- Keyboard shortcuts for frequent actions in a software application.

8. Aesthetic and Minimalist Design

Definition: Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

Example:

- A clean, uncluttered homepage with a clear focus on essential content and actions.

9. Help Users Recognize, Diagnose, and Recover from Errors

Definition: Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

Example:

- An error message that explains why a password was rejected and offers tips for creating a valid one.

10. Help and Documentation

Definition: Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

Example:

- An online help center with searchable FAQs and step-by-step guides.